

OnRamps Precalculus
Final Exam Review 2024-2025

Learning Targets

Unit 4A

- 1) Use basic trigonometry to solve right triangles.
- 2) Understand and use the reciprocal, quotient, and Pythagorean identities.
- 3) Construct the unit circle using the relationships found in special right triangles.
- 4) Know the sign of the trigonometric values in each quadrant.
- 5) Use trigonometric identities to simplify trigonometric expressions.
- 6) Solve trigonometric equations by using trigonometric identities and algebraic techniques.
- 7) Calculate trigonometric values from angles found on the unit circle.
- 8) Determine trigonometric ratios based on a given trigonometric value in a specified quadrant.
- 9) Graph the parent trigonometric functions: sine, cosine, tangent.

Unit 4B

- 1) Graph and determine transformations of trigonometric functions from an equation, including key points.
- 2) Know the domain and range of inverse trigonometric functions to evaluate inverse trig expressions.
- 3) Use knowledge of inverse trigonometric functions to simplify composite trig expressions.
- 4) Write an equation for a trigonometric function given key points from the graph.

Unit 5

- 1) Identify the intercepts, end behavior, and discontinuities of rational functions.
- 2) Use attributes and graphs of rational functions to write the equation.
- 3) Determine the limit of a function using graphs and tables.
- 4) Apply the power rule to differentiate equations.
- 5) Evaluate instantaneous rate of change at a given x -value.
- 6) Use derivatives to determine the increasing and decreasing intervals and the relative extrema of functions.

Unit 6

- 1) Graph parametric equations.
- 2) Convert between rectangular relations and parametric equations.
- 3) Create parametric equations to model real-world problems.
- 4) Determine the component form and magnitude given a vector on a plane.
- 5) Determine the magnitude and direction of a vector given in component form.
- 6) Graph and label given points in the polar coordinate system.
- 7) Convert between rectangular and polar coordinates.
- 8) Graph polar equations.
- 9) Write equations of polar functions given a graph.

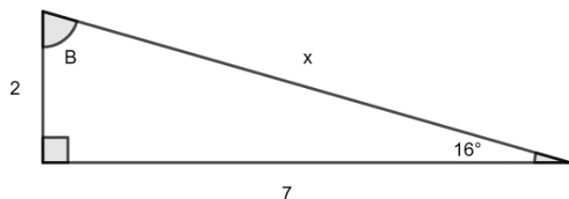
Additional Notes

- Anything not listed in the learning targets will NOT be tested on the Final Exam.
- A calculator will NOT be allowed.
- Notes or other outside resources will NOT be allowed.
- You will have 45 minutes to complete the Final Exam.
- The Final Exam will be only one part; there are no open-response questions.
- Students should have the ratio, reciprocal, and Pythagorean identities memorized, as well as the Unit Circle.

Final Exam Practice Questions

Unit 4A

1. Find the length x and measure of $\angle B$



$$x = \underline{\hspace{2cm}}$$

$$\angle B = \underline{\hspace{2cm}}$$

2. Evaluate each trig expression.

a) $\cos\left(\frac{\pi}{3}\right) = \underline{\hspace{2cm}}$

b) $\tan\left(\frac{5\pi}{6}\right) = \underline{\hspace{2cm}}$

c) $\sin(\pi) = \underline{\hspace{2cm}}$

d) $\csc\left(\frac{2\pi}{3}\right) = \underline{\hspace{2cm}}$

e) $\sec\left(\frac{5\pi}{4}\right) = \underline{\hspace{2cm}}$

f) $\cot\left(\frac{7\pi}{3}\right) = \underline{\hspace{2cm}}$

3. Given $\sin \theta = -\frac{1}{3}$, with $\frac{3\pi}{2} < \theta < 2\pi$, determine the following. Rationalize the denominator if necessary.

a) $\cos(\theta) = \underline{\hspace{2cm}}$

b) $\tan(\theta) = \underline{\hspace{2cm}}$

c) $\sec(\theta) = \underline{\hspace{2cm}}$

d) $\csc(\theta) = \underline{\hspace{2cm}}$

e) $\cot(\theta) = \underline{\hspace{2cm}}$

4. Given $\tan \theta = \frac{5}{12}$, with $\pi < \theta < \frac{3\pi}{2}$, determine the following. Rationalize the denominator if necessary.

a) $\sin(\theta) = \underline{\hspace{2cm}}$

b) $\cos(\theta) = \underline{\hspace{2cm}}$

c) $\sec(\theta) = \underline{\hspace{2cm}}$

d) $\csc(\theta) = \underline{\hspace{2cm}}$

e) $\cot(\theta) = \underline{\hspace{2cm}}$

5. Solve for $0 \leq \theta < 2\pi$.

a) $2 \sin \theta + 1 = 0$

b) $\tan^2 \theta - 3 = 0$

c) $\sin(3\theta) = -\frac{\sqrt{2}}{2}$

d) $\cos^2(2\theta) = \frac{1}{4}$

6. Solve for $0 \leq \theta < 2\pi$.

a) $\sin \theta + \sin \theta \tan \theta = 0$

b) $2 \cos \theta = \cos \theta \csc \theta$

c) $2 \cos^2 \theta - \cos \theta = 1$

d) $\sin^2 \theta + \sin \theta \cot \theta = 1$

7. Fully simplify each trig expression.

a) $\frac{\tan \theta + \cot \theta}{\csc \theta}$

b) $\sec \theta (\sec \theta - \cos \theta)$

c) $\tan \theta \csc \theta - \tan^2 \theta \cos \theta$

d) $\cos \theta \csc \theta (\sec^2 \theta - 1)$

8. Verify the following identities.

a) $\sin^2 \theta - \cos^2 \theta = 2 \sin^2 \theta - 1$

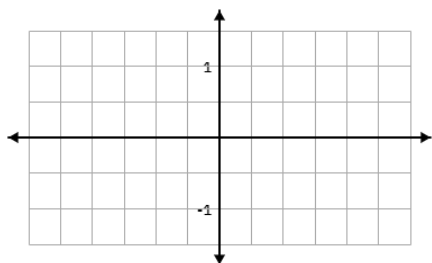
b) $\frac{\cos^2 \theta}{1 + \sin \theta} = 1 - \sin \theta$

c) $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$

d) $\frac{\cos^4 \theta - \sin^4 \theta}{\cos^2 \theta \sin^2 \theta} = \csc^2 \theta - \sec^2 \theta$

9. On each grid provided, draw the graph of $\sin x$, $\cos x$, and $\tan x$. Label each horizontal gridline to be $\frac{\pi}{2}$ units apart. Label the domain and range of each. Remember to draw the asymptotes for the tangent function.

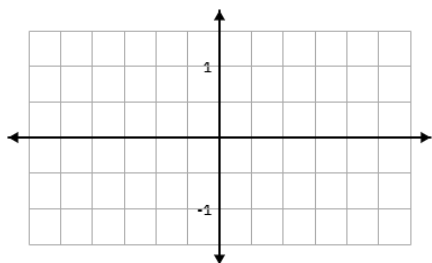
$y = \sin x$



D: _____

R: _____

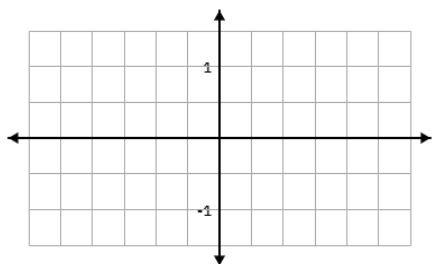
$y = \cos x$



D: _____

R: _____

$y = \tan x$



D: _____

R: _____

Unit 4B

1. Describe the each of the indicated characteristics of the function $f(x) = 4 + 6 \sin(3x + \pi)$.

a) Transformations: _____

b) Period: _____

c) The y -intercept is at _____

d) Minimum y -value is _____ and maximum y -value is _____

2. Evaluate each inverse trig expression.

a) $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right) =$ _____

b) $\sin^{-1}\left(\frac{1}{2}\right) =$ _____

c) $\tan^{-1}(-\sqrt{3}) =$ _____

3. Evaluate each composite expression.

a) $\sin^{-1}\left(\sin\left(\frac{3\pi}{2}\right)\right) =$ _____

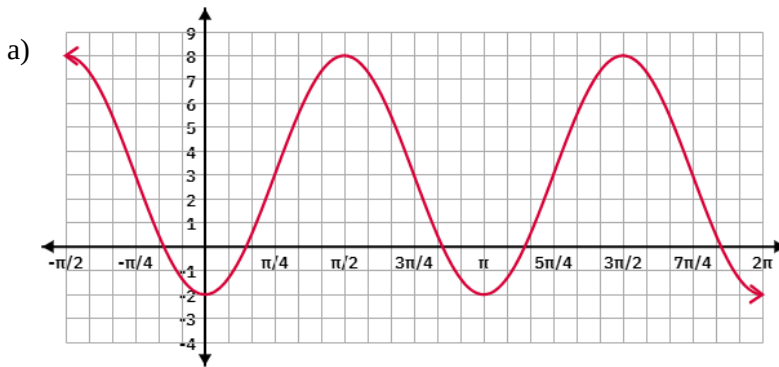
b) $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right) =$ _____

c) $\cos^{-1}\left(\sin\left(\frac{4\pi}{3}\right)\right) =$ _____

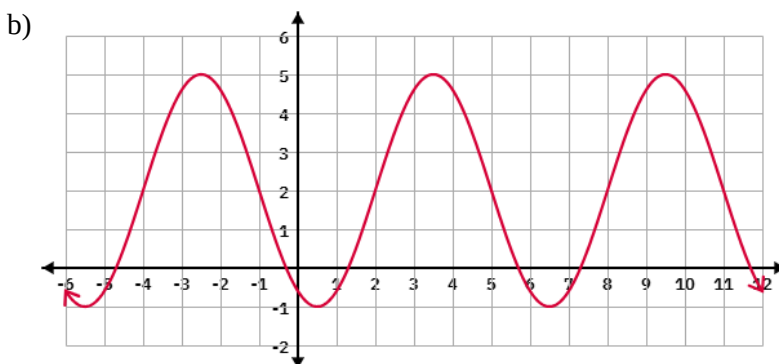
d) $\sin^{-1}\left(\cos\left(\frac{11\pi}{6}\right)\right) =$ _____

e) $\tan^{-1}(\cos(\pi)) =$ _____

4. Write a sinusoidal function that accurately describes the graph given below. You may use either sine or cosine.



$f(x) =$ _____



$f(x) =$ _____

Unit 5

1. Identify the major attributes of each rational function. That is, give the x - and y -intercepts, vertical and horizontal asymptotes, removable discontinuities (holes), describe the end behavior of the function and its domain and range.

a) $f(x) = \frac{x-6}{x-3}$

b) $g(x) = \frac{x^2-25}{x+5}$

c) $h(x) = \frac{x^2+x-6}{2x^2-8}$

d) $j(x) = \frac{x-1}{x^2-2x+1}$

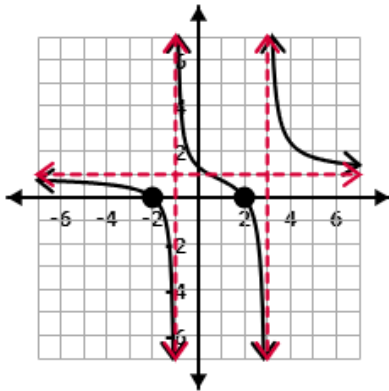
e) $k(x) = \frac{6x^2+7x-3}{3x^2-13x+4}$

**I put this one on here because you have to be careful factoring and work with fractions*

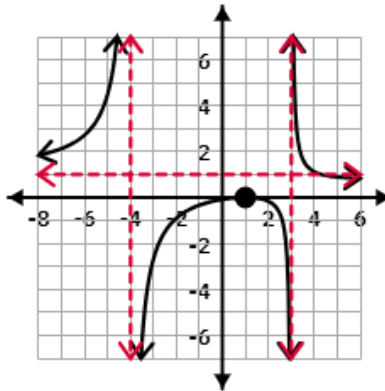
2. Create a rational function of the form $f(x) = \frac{p(x)}{q(x)}$ that satisfies the following:

- x -intercept at $(-4, 0)$
- Vertical asymptote at $x = 5$
- Hole at $x = -1$
- Horizontal asymptote at $y = \frac{3}{2}$

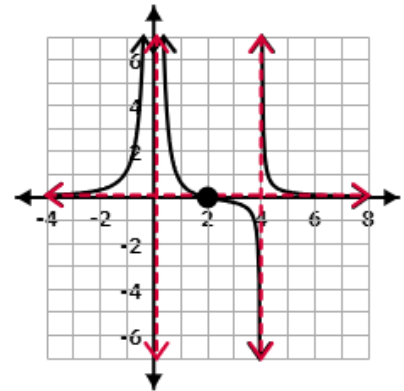
3. Write a rational function in factored form that might generate each of the graphs below.



$f(x) = \underline{\hspace{2cm}}$



$f(x) = \underline{\hspace{2cm}}$



$f(x) = \underline{\hspace{2cm}}$

4. Evaluate the following limits given the graph of $f(x) = \frac{x^2+6x+5}{x^2-1}$.

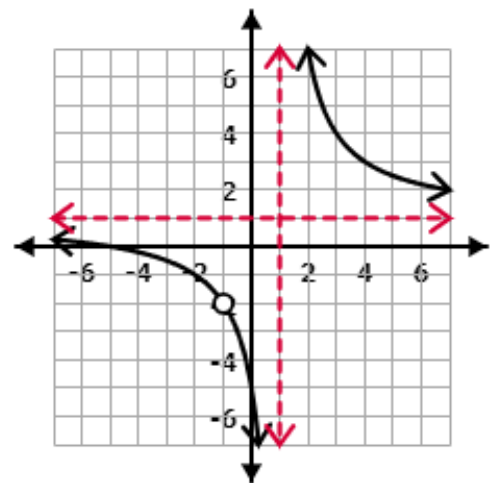
a) $\lim_{x \rightarrow 4} f(x) = \underline{\hspace{2cm}}$

b) $\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}}$

c) $\lim_{x \rightarrow -1} f(x) = \underline{\hspace{2cm}}$

d) $\lim_{x \rightarrow -5} f(x) = \underline{\hspace{2cm}}$

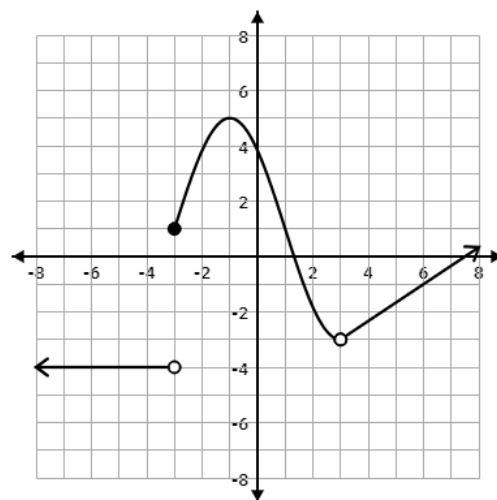
e) $\lim_{x \rightarrow \infty} f(x) = \underline{\hspace{2cm}}$



5. Based off the graph of $f(x)$ shown, evaluate the following.

Write “DNE” if the limit does not exist.

- a) $\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}}$
 b) $\lim_{x \rightarrow \infty} f(x) = \underline{\hspace{2cm}}$
 c) $\lim_{x \rightarrow -3} f(x) = \underline{\hspace{2cm}}$
 d) $\lim_{x \rightarrow 3} f(x) = \underline{\hspace{2cm}}$



6. Find the derivative of each function.

- a) $f(x) = \frac{1}{6}x^2 - 3x + 7$
 b) $g(x) = \sqrt[3]{x} + x$
 c) $h(x) = \frac{4}{x^2} - \frac{1}{x}$

7. Answer the following given $p(x) = 4\sqrt{x} + 1$.

- a) What is the average rate of change of $p(x)$ over the interval $[0, 9]$?
**Not said to be tested, still good review.*
 b) What is the instantaneous rate of change of $p(x)$ at $x = 4$?

8. Answer the following given $f(x) = x^3 - 6x^2 + 5$.

- a) What is the instantaneous rate of change of $f(x)$ at $x = -2$?
 b) Over what interval is f increasing?
 c) Over what interval is f decreasing?
 d) Identify all extrema and classify each as a relative maximum or minimum value.

9. Answer the following given $f(x) = x - 2\sqrt{x}$.

- a) Over what interval is f increasing?
 b) Over what interval is f decreasing?
 c) Identify all extrema and classify each as a relative maximum or minimum value.

Unit 6

1. Answer the following given the parametric function below.

$$\begin{cases} x(t) = t^3 - 2t \\ y(t) = t^2 + t \end{cases}$$

- Create a table that gives $x(t)$ and $y(t)$ for values of $t = \{-2, -1, 0, 1, 2\}$.
- Using the table you wrote, draw the graph generated by this function for $-2 \leq t \leq 2$ on the xy -plane.

2. Answer the following given the parametric function below. (Calculator required – practicing part b is the more important part of this problem).

$$\begin{cases} x(t) = \cos(3t) \\ y(t) = \sin(2t) \end{cases}$$

- Create a table that gives $x(t)$ and $y(t)$ for values of $t = \{0, \pi/6, \pi/4, \pi/3, \pi/2\}$. Use a calculator to give a decimal approximation for each value.
- Using the table you wrote, draw the graph generated by this function for $0 \leq t \leq \frac{\pi}{2}$ on the xy -plane.
- *For fun: this is called a Lissajous curve. You can see the full curve in Desmos if you graph it for $0 \leq t \leq 2\pi$.

3. A boat is spotted 15 miles south and 7 miles west of a port in San Pedro at 8:00 am. At 10:00 am, the boat has moved to a location 8 miles south and 3 miles east of the port.

- How far did the boat travel in the first hour?
- At noon (12:00 pm), where is the boat in relation to the port?
- Write a parametric function that will give the boat's position (in relation to the port) at any time.
- Predict the boat's location relative to the port at 4:00 pm. Assume the boat will simply continue in its path regardless of whether it hits land – this boat stops for no one.

4. You are watching a race car travel in an elliptical path. The race car starts at the co-vertex of the ellipse furthest from you and travels counterclockwise around the track.

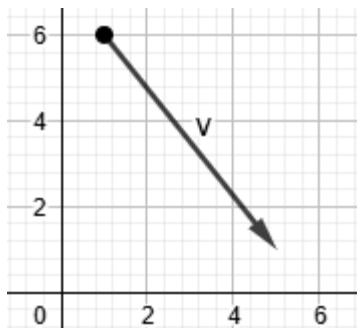
- The race car starts at a distance 1520 ft north and 2740 ft east of your position.
 - After 15 seconds, the car has reached the vertex of the elliptical path, 860 ft north and 100 ft east of your position.
 - After another 15 seconds, the car has reached the other co-vertex of the ellipse, opposite of its starting point, located 200 ft north and 2740 ft east of your position.
 - Assume the race car is able to reach each vertex/co-vertex in exactly 15 seconds, taking one minute exactly to finish one lap.
- Write the rectangular equation that models the path of the race car in relation to you.
 - Write a parametric function that models the path of the race car. Be sure to consider the race cars starting point and motion around the path.

***It is much more likely you are asked a question about parametric applications that looks like #3 than #4. Still, you want to be prepared in using sine/cosine parametric functions should they come up.**

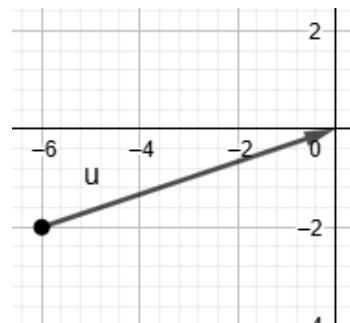
5. Rewrite the rectangular equation (given below) as a system of parametric equations, $x(t)$ and $y(t)$.

$$\frac{(x+3)^2}{100} + \frac{(y-1)^2}{16} = 1$$

6. Answer the following given the image of vectors $\vec{v}$ and $\vec{u}$.



It is a little hard to see, but vector $\vec{v}$ ends at (5, 1).

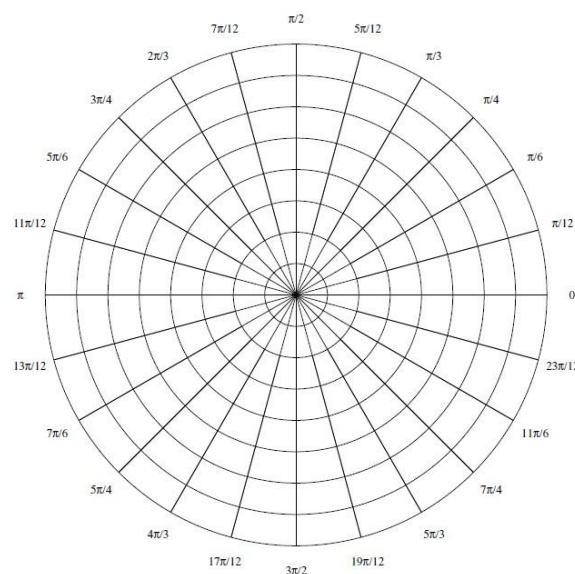


- Find the component form of $\vec{u}$ and $\vec{v}$.
 - Find the direction of $\vec{u}$ and $\vec{v}$ (center each vector at the origin and find $\theta_{\vec{u}}$ and $\theta_{\vec{v}}$).
 - Write the component form of $4\vec{u} - 3\vec{v}$.
7. Given each vector, find its magnitude and direction.

- $\vec{u} = \langle 10\sqrt{3}, 10 \rangle$
- $\vec{v} = \langle 2\sqrt{2}, -2\sqrt{6} \rangle$
- $\vec{w} = \langle -4, 4 \rangle$

8. Graph each coordinate (r, θ) on the polar coordinate system given.

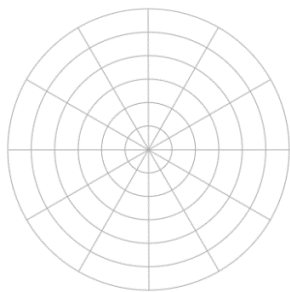
- $\left(3, \frac{\pi}{4}\right)$
- $\left(5, -\frac{2\pi}{3}\right)$
- $\left(-2, \frac{\pi}{2}\right)$
- $\left(-4, -\frac{\pi}{6}\right)$
- $(4, 0)$
- $(0, 4)$



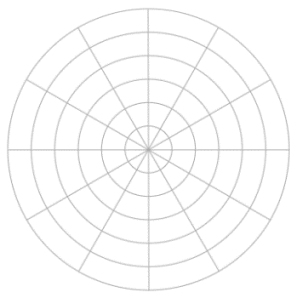
9. Convert between rectangular coordinates and polar coordinates.

(x, y)	(r, θ)
	$\left(4, \frac{3\pi}{4}\right)$
$(-\sqrt{3}, -1)$	
	$(-3, \pi)$
$(0, 5)$	

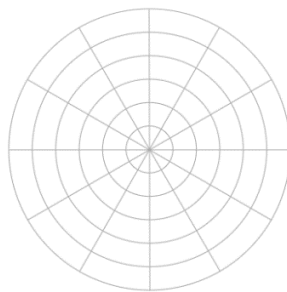
10. Sketch each polar curve given the equation.



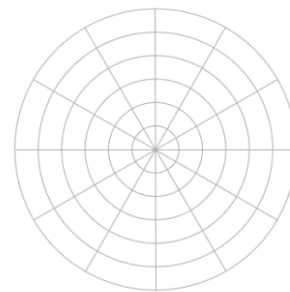
a) $r = 5 \cos \theta$



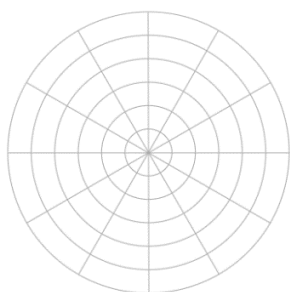
b) $r = 2$



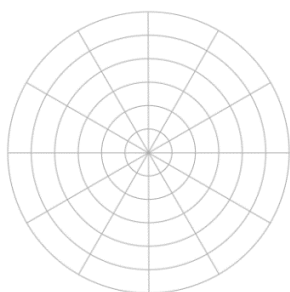
c) $\theta = 5\pi/6$



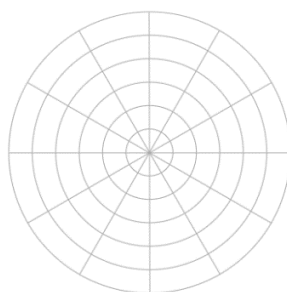
d) $r = -3 \sin \theta$



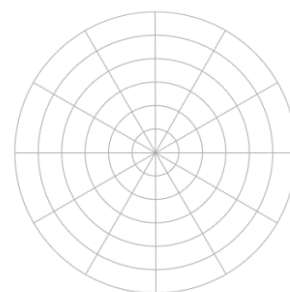
e) $r = 2 - 3 \cos \theta$



f) $r = 6 \cos 2\theta$

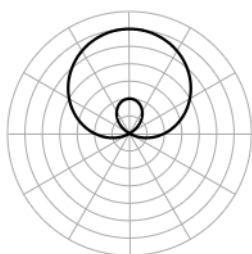


g) $r = 3 + 3 \sin(\theta)$

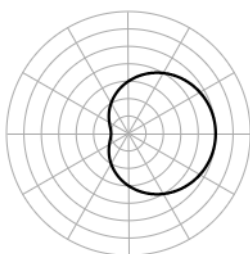


h) $r = 4 - \sin \theta$

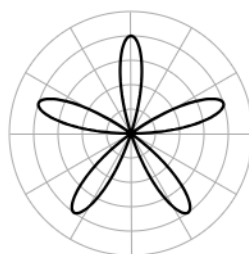
11. Write the polar equation given the graph of each curve.



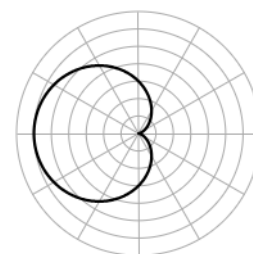
a) _____



b) _____



c) _____



d) _____